GRB Research Paper

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1 Introduction

2 GRB observations

2.1 GRB080916C

The GRB was observed on September 16, 2008, at 00:12:45.613542 UT (T_0) by the Fermi Gamma-ray Space Telescope. Detected by both the Gamma-ray Burst Monitor (GBM) and the Large Area Telescope (LAT) instruments onboard Fermi (Goldstein & van der Horst 2008; Tajima et al. 2008), it is one of the most energetic GRBs to date. The burst exhibited a complex temporal structure with multiple peaks and a duration of approximately 66 seconds in the GBM energy range. The LAT detected high-energy photons up to 13 GeV, providing valuable insights into the emission mechanisms of GRBs (Abdo et al. 2009). The preliminary analysis performed by LAT, the InterPlanetary Network (IPN), and Swift provided an initial localization of the burst around $\langle \mathrm{RA}, \mathrm{Dec} \rangle = \langle 120°, -55° \rangle$ (J2000).

The redshift of GRB 080916C was determined to be 4.35 through spectroscopic observations of its afterglow (Greiner et al. 2009), indicating that it occurred when the universe was less than 1.5 billion years old. The afterglow for GRB080916C was observed across multiple wavelengths, including X-ray, optical, and radio, providing a comprehensive view of the burst's evolution over time (Clemens et al. 2008a,b; Greiner et al. 2009; Hoversten et al. 2008; Kennea 2008; Nagayama 2008; Perri et al. 2008).

2.2 GRB110721A

The GRB was detected on July 21, 2011, at 04:47:43.75 UT (T_0) by the Fermi Gamma-ray Space Telescope. The burst was observed by both the GBM and LAT instruments onboard Fermi (Tierney & von Kienlin 2011; Vasileiou et al. 2011). The burst exhibited a fast rise, exponential decay curve of ~24 s in the GBM energy range. The target of opportunity (ToO) observations by Swift began approximately 26 h after the initial GBM trigger time (Greiner et al. 2011). Preliminary analysis provided an initial localization of the burst around $\langle \mathrm{RA}, \mathrm{Dec} \rangle = \langle 333.4°, -39° \rangle$ (J2000) (Greiner et al. 2011; Vasileiou et al. 2011).

The redshift of GRB110721A was determined to be 0.382 through spectroscopic observations of its afterglow (Berger 2011). The source being a low-luminosity AGN was later confirmed by the Swift team (Grupe et al. 2011) with a 0.3 keV to 10 keV flux of 2×10^{43} erg s^{-1}.

2.3 GRB110731A

GRB 110731A was detected on July 31, 2011, at 11:09:30 UT (T_0) by both the *Swift*-BAT and the *Fermi* observatory. BAT provided an initial localization at $\langle 280.521°, -28.546° \rangle$ with a $3'$ uncertainty, and the prompt emission showed a multi-peaked structure lasting about 20 s (Oates et al. 2011). XRT observations began 66 s after the trigger and identified a bright uncatalogued X-ray source, while UVOT detected an optical afterglow with an initial magnitude of 15.8 in the white filter (Oates et al. 2011). Ground-based observations with the Faulkes Telescopes confirmed a fading optical counterpart with $m_R \simeq 16.4$ at 28 minutes after the burst (Bersier 2011).

A refined BAT analysis yielded a duration of $T_{90} = 38.8 \pm 13.0$ s and a time-averaged spectrum described by a power law with photon index $\Gamma = 1.15 \pm 0.05$ (Krimm et al. 2011). At higher energies, the *Fermi*-LAT detected emission above 100 MeV with more than 10σ significance from a position consistent with the *Swift* localization (Bregeon et al. 2011). The *Fermi*-GBM measured a single dominant pulse with $T_{90} = 7.3 \pm 0.3$ s a cutoff power-law spectrum with $E_{\rm peak} \simeq 317$ keV (Gruber 2011).

Optical spectroscopy with Gemini-N/GMOS revealed a broad Lyα absorption feature and several metal lines, establishing a redshift of $z = 2.83$. The acquisition image yielded $R \simeq 22.0$, consistent with a fading afterglow phenomena (Tanvir et al. 2011).

2.4 GRB150210A

GRB 150210A was detected on 10 February 2015 at 22:26:24 UT (T_0) by the *Fermi* Gamma-ray Space Telescope. The burst was observed by both the GBM and LAT instruments, with the GBM localizing it to $\langle \mathrm{RA}, \mathrm{Dec} \rangle \simeq \langle 112.9°, +12.4° \rangle$ (J2000) and measuring a single FRED-like pulse of duration $T_{90} \simeq 31$ s in the 50 keV to 300 keV band (Burns 2015). The time-averaged GBM spectrum is described by a cutoff power law with photon index $\alpha = -1.05 \pm 0.01$ and peak energy $E_{\rm p} = 3118 \pm 216$ keV, yielding a fluence of $(1.35 \pm 0.01) \times 10^{-5}\,\mathrm{erg\,cm^{-2}}$ in the 10 keV to 1000 keV band (Burns 2015).

Fermi-LAT detected high-energy emission with an on-ground localization of $\langle 112.15°, 13.27° \rangle$ (J2000) and a 90% containment radius of 0.33°. More than twenty photons above 100 MeV were detected in the first 100 s, including a ~ 1 GeV photon at $T_0 + 2$ s (Zhu et al. 2015).

Konus–Wind also detected the burst, finding a bright multi-peaked pulse with emission up to ~ 10 MeV and measuring a fluence of $(7.8^{+0.9}_{-0.8}) \times 10^{-5}\,\mathrm{erg\,cm^{-2}}$ in the 20 keV to 10 000 keV band (Golenetskii et al. 2015).

Swift-XRT performed tiled follow-up observations between $T_0 + 66.9$ and $T_0 + 95.4$ ks, detecting no uncatalogued X-ray sources and placing 3σ flux limits of $(1.5\text{–}3.3) \times 10^{-13}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$ (Mangano et al. 2015). MASTER optical imaging identified a transient of magnitude $m \approx 17.4$ unrelated to the burst, likely a supernova (Gres et al. 2015).

3 Data preparation and analysis

For the analysis, we used the data from the Fermi GBM and LAT for our spectral analysis, covering an energy range from 10 keV to 100 GeV. The data was downloaded from the HEASARC servers for GBM, and Fermi Science Support Center (FSSC) for LAT.

3.1 GBM data

In the data selection, the energy range for all NaI detectors was taken from 10 keV to 900 keV, and for BGO it was 0.25 MeV to 40 MeV. The detectors were selected with the brightest signal and the smallest angular separation from the source. The positional discrepancies were removed from the GBM/LAT data using the positions provided by Ajello et al. (2019). For this purpose we created new response files (RSP) and RSP2 files using GBMRSP v2.0 and GBM DRM database v2.0. For the T_{90} duration, the values provided by Ajello et al. (2019) were slightly tweaked to accommodate the GBM time resolution of 0.064 s. The background was modeled via a polynomial of degree $1 - 4$ for each detector.

For this study, some episodes were selected with non-standard durations extending beyond the T_{90} interval. These were chosen to accommodate the possible emission missed by the T_{90} window, either at the onset or in the fading tail of the GRB emission. The details for time-integrated analysis duration are given in table 1.

3.1.1 GRB080916C

From the NaI detectors, n3 and n4 were selected with angular separations of 19° and 41° respectively. For the BGO detectors, b0 was chosen, with an angular separation of 67°. The time-integrated analysis was performed for the T_{90} duration of 1.280 s to 64.256 s.

3.1.2 GRB110721A

From the NaI detectors, n6, n7, n9 were selected with angular separation of 29°, 32° and 29° respectively. For the BGO detectors, b1 was chosen with an angular separation of 50°. The time-integrated analysis was performed for the T_{90} duration of 0.000 s to 21.824 s.

3.1.3 GRB110731A

From the NaI detectors, n0, n3, n6 were selected with angular separation of 19°, 41° and 21° respectively. For the BGO detectors, b0 was chosen with an angular separation of 86°. The time-integrated analysis was performed for the T_{90} duration of 0.064 s to 7.488 s.

3.1.4 GRB150210A

From the NaI detectors, n9, na, and nb were selected with angular separations of 13°, 43°, and 48°, respectively. For the BGO detectors, b1 was chosen, with an angular separation of 46°. The time-integrated analysis was performed for the T_{90} duration of 0.064 s to 31.360 s.

4 Physics calculations

see, `https://www.aanda.org/articles/aa/pdf/2023/05/aa45414-22.pdf` this paper has physics on both Amati and Yonetoku relationship, can be good.

4.1 Yonetoku relationship

see, `https://www.aanda.org/articles/aa/pdf/2023/05/aa45414-22.pdf`
see, `https://arxiv.org/pdf/1611.05732`

Table 1: Time-integrated and time-resolved duration for the selected GRBs. The EX episodes are outside the standard T_{90} duration, taken to accommodate excess on-set or fading tail release of GRBs.

Episode	GRB080916C	GRB110721A	GRB110731A	GRB150210A
Time-integrated				
T_{90}	1.280 s – 64.256 s	0.000 s – 21.824 s	0.064 s – 7.448 s	0.064 s – 31.360 s
Time-resolved				
EX–A	−0.128 s – 4.864 s	−0.384 s – 1.344 s	−0.512 s – 2.112 s	−0.384 s – 0.832 s
I	1.280 s – 4.864 s	0.000 s – 1.344 s	0.064 s – 2.112 s	0.064 s – 0.832 s
II	4.864 s – 15.040 s	1.344 s – 21.824 s	2.112 s – 5.184 s	0.832 s – 2.176 s
III	15.040 s – 55.296 s	–	5.184 s – 7.488 s	2.176 s – 3.328 s
IV	55.296 s – 60.992 s	–	–	3.328 s – 17.344 s
V	60.992 s – 64.256 s	–	–	17.344 s – 31.36 s
EX–B	60.992 s – 67.904 s	–	5.184 s – 9.984 s	–

see, https://www.nature.com/articles/s41467-019-09281-z
ToBeDecided

4.2 Amati relationship

In our sample, three out of four GRBs have measured redshifts, GRB080916C/110721A/110731A. GRB150210A doesn't have a measured redshift.

See, https://www.scirp.org/pdf/IJAA_2016110915245229.pdf eq 1 and eq 2 ToBeDecided

4.3 Isotropic energy calculations

ToBeDecided

4.4 gamma-gamma opacity calculations

ToBeDecided

4.5 Lorentz factor calculations

ToBeDecided

4.6 hardness-to-softness ratio

Have to look this up as well, i think this is a thing
ToBeDecided

References

Abdo, A. A., Ackermann, M., Arimoto, M., et al. 2009, Science, 323, 1688

Ajello, M., Arimoto, M., Axelsson, M., et al. 2019, ApJ, 878, 52

Berger, E. 2011, GRB Coordinates Network, 12193, 1

Bersier, D. 2011, GRB Coordinates Network, 12216, 1

Bregeon, J., McEnery, J., & Ohno, M. 2011, GRB Coordinates Network, 12218, 1

Burns, E. 2015, GRB Coordinates Network, 17432, 1

Clemens, C., Rossi, A., Greiner, J., et al. 2008a, GRB Coordinates Network, 8272, 1

Clemens, C., Rossi, A., Greiner, J., et al. 2008b, GRB Coordinates Network, 8273, 1

Goldstein, A. & van der Horst, A. 2008, GRB Coordinates Network, 8245, 1

Golenetskii, S., Aptekar, R., Frederiks, D., et al. 2015, GRB Coordinates Network, 17440, 1

Greiner, J., Clemens, C., Krühler, T., et al. 2009, A&A, 498, 89

Greiner, J., Updike, A. C., Kruehler, T., & Sudilovsky, V. 2011, GRB Coordinates Network, 12192, 1

Gres, O., Ivanov, K., Yazev, S., et al. 2015, GRB Coordinates Network, 17446, 1

Gruber, D. 2011, GRB Coordinates Network, 12221, 1

Grupe, D., Swenson, C. A., Racusin, J. L., & Wolf, C. 2011, GRB Coordinates Network, 12212, 1

Hoversten, E. A., Schady, P., & Perri, M. 2008, GRB Coordinates Network, 8262, 1

Kennea, J. A. 2008, GRB Coordinates Network, 8253, 1

Krimm, H. A., Barthelmy, S. D., Baumgartner, W. H., et al. 2011, GRB Coordinates Network, 12217, 1

Mangano, V., Stroh, M. C., Gompertz, B. P., et al. 2015, GRB Coordinates Network, 17453, 1

Nagayama, T. 2008, GRB Coordinates Network, 8274, 1

Oates, S. R., Beardmore, A. P., Holland, S. T., et al. 2011, GRB Coordinates Network, 12215, 1

Perri, M., Preger, B., & Stratta, G. 2008, GRB Coordinates Network, 8261, 1

Tajima, H., Bregeon, J., Chiang, J., & Thayer, G. 2008, GRB Coordinates Network, 8246, 1

Tanvir, N. R., Wiersema, K., Levan, A. J., Cenko, S. B., & Geballe, T. 2011, GRB Coordinates Network, 12225, 1

Tierney, D. & von Kienlin, A. 2011, GRB Coordinates Network, 12187, 1

Vasileiou, V., Piron, F., Tierney, D., et al. 2011, GRB Coordinates Network, 12188, 1

Zhu, S., Bissaldi, E., Racusin, J., et al. 2015, GRB Coordinates Network, 17438, 1

A GRB parameters

A.1 GRB080916C

Table 2: Fitted parameters for time-integrated and time-resolved spectral analysis of GRB080916C. The values in parenthesis are the uncertainty in the measured parameter. The BEST models are highlighted with a light gray strip. The marked parameters in red indicate the values that are just beyond the error criteria limit. The percentage above threshold is indicated in superscript for reference. The models not included here either did not converge, or produced errors much higher than the threshold criteria.

GRB080916C									
	SBPL/Band/CPL				PL		BB		
	Amplitude [$\times 10^{-3}$] [ph cm^{-2} s^{-1} keV^{-1}]	α	β	Break/Peak Energy [keV]	Amplitude [$\times 10^{-3}$] [ph cm^{-2} s^{-1} keV^{-1}]	α	Amplitude [$\times 10^{-6}$] [ph cm^{-2} s^{-1} keV^{-1}]	kT [keV]	C-Stat/DOF
Time integrated: 1.280 s − 64.256 s									
PL	–	–	–	–	8.174(0.064)	−1.774(0.005)	–	–	9528.2715/367
PL+BB	–	–	–	–	4.175(0.067)	−1.874(0.007)	4.661(0.177)	59.489(0.684)	1610.6810/365
SBPL	14.504(0.146)	−1.096(0.012)	−2.196(0.011)	271.584(12.862)	–	–	–	–	931.2349/365
SBPL+BB	10.461(0.415)	−1.291(0.025)	−2.251(0.023)	701.613(5.599)	–	–	3.845(0.537)	44.452(2.052)	892.9536/363
Band	17.482(0.363)	−1.016(0.016)	−2.202(0.012)	512.213(27.027)	–	–	–	–	947.1155/365
Band+BB	11.127(0.529)	−1.263(0.028)	−2.270(0.028)	1392.777(275.484)	–	–	3.277(0.485)	46.841(2.024)	897.5037/363
CPL	16.996(0.030)	−1.033(0.014)	–	571.760(26.575)	–	–	–	–	2303.7300/366
CPL+PL	16.563(0.396)	−0.699(0.058)	–	439.416(19.981)	2.069(0.202)	−1.822(0.013)	–	–	1009.7907/364
CPL+PL+BB	8.492(0.083)	−0.746(0.122)	–	797.341(94.660)	2.711(0.264)	−1.854(0.013)	7.265(1.727)	37.251(2.268)	949.1119/362
Episode EX–A: −0.128 s − 4.864 s									
PL	–	–	–	–	20.932(0.282)	−1.616(0.011)	–	–	3735.2476/347
PL+BB	–	–	–	–	7.084(0.378)	−1.801(0.028)	8.312(0.048)	69.446(1.207)	821.0788/345
SBPL	33.812(0.635)	−0.811(0.027)	−2.334(0.047)	291.658(19.468)	–	–	–	–	450.5526/345
SBPL+BB	22.067(1.179)	−1.015(0.037)	−2.510(0.067)	782.033(145.414)	–	–	14.608(3.080)	41.205(3.144)	412.6351/343
Band	42.874(1.507)	−0.705(0.033)	−2.372(0.050)	533.223(34.382)	–	–	–	–	444.8021/345
Band+BB	25.084(1.780)	−0.954(0.045)	−2.546(0.077)	1308.992(247.864)	–	–	12.323(2.765)	42.493(3.269)	414.2776/343
CPL	40.358(1.167)	−0.754(0.023)	–	618.369(36.395)	–	–	–	–	564.4187/346
CPL+BB	23.849(1.219)	−0.984(0.033)	–	1576.787(222.279)	–	–	12.553(2.581)	43.256(2.945)	515.4531/344
CPL+PL+BB	18.956(3.065)	−0.727(0.151)	–	1272.322(188.478)	2.720(0.923)	−1.817(0.051)	18.775(5.263)	40.071(2.700)	415.4321/342
Episode I: 1.280 s − 4.864 s									
PL	–	–	–	–	23.702(0.344)	−1.612(0.011)	–	–	3167.8596/347
PL+BB	–	–	–	–	8.413(0.453)	−1.779(0.028)	8.264(0.533)	71.874(1.379)	795.0419/345
SBPL	37.1670(0.7514)	−0.835(0.028)	−2.307(0.047)	311.673(23.187)	–	–	–	–	443.6357/345
SBPL+BB	25.022(1.302)	−1.004(0.038)	−2.477(0.067)	796.206(147.815)	–	–	18.514(4.567)	38.935(3.341)	405.1061/343
Band	46.241(1.688)	−0.733(0.034)	−2.341(0.051)	577.109(41.998)	–	–	–	–	435.5710/345
Band+BB	28.639(1.999)	−0.938(0.045)	−2.506(0.076)	1330.902(250.387)	–	–	15.714(4.213)	39.756(3.588)	406.6308/343
CPL	43.480(1.301)	−0.784(0.030)	–	680.654(45.621)	–	–	–	–	564.0950/346
Episode II: 4.864 s − 15.040 s									
PL	–	–	–	–	11.986(0.167)	−1.734(0.008)	–	–	2265.7590/356
PL+BB	–	–	–	–	7.081(0.184)	−1.803(0.010)	3.894(0.330)	66.869(1.720)	693.2336/354
SBPL	18.442(0.344)	−1.196(0.020)	−2.106(0.021)	394.640(49.756)	–	–	–	–	471.1628/354
SBPL+BB	14.627(0.621)	−1.325(0.027)	−2.209(0.047)	1313.309(428.740)	–	–	3.879(1.281)	44.682(5.077)	459.0358/352
Band	20.289(0.642)	−1.152(0.025)	−2.122(0.024)	885.708(129.277)	–	–	–	–	470.5969/354
Band+BB	15.573(0.837)	−1.292(0.033)	−2.217(0.049)	2438.760(739.350)	–	–	3.170(1.193)	45.966(5.700)	459.2626/352
CPL	19.679(0.518)	−1.174(0.022)	–	1084.105(148.421)	–	–	–	–	1246.1093/355
CPL+PL	17.543(0.851)	−0.912(0.110)	–	693.261(95.282)	3.324(0.791)	−1.737(0.031)	–	–	511.8681/353
Episode III: 15.040 s − 55.296 s									
PL	–	–	–	–	7.037(0.079)	−1.789(0.007)	–	–	4914.1157/367
PL+BB	–	–	–	–	3.614(0.082)	−1.897(0.009)	4.401(0.236)	57.785(0.941)	928.7520/365
SBPL	12.710(0.179)	−1.115(0.017)	−2.215(0.016)	258.482(17.188)	–	–	–	–	696.9452/365
SBPL+PL	11.284(0.445)	−0.631(0.109)	−2.245(0.064)	179.595(15.411)	1.907(0.381)	−1.971(0.073)	–	–	668.3735/363
SBPL+BB	8.810(0.626)	−1.344(0.043)	−2.234(0.031)	597.467(192.102)	–	–	2.803(0.586)	49.141(2.605)	672.8546/363
SBPL+PL+BB	7.088(1.158)	−0.906(0.230)	−2.341(0.095)	365.924(135.736)	1.947(0.462)	−1.929(0.044)	5.473(2.166)	37.947(5.770)	663.9378/361
Band	15.587(0.477)	−1.027(0.022)	−2.217(0.017)	465.404(34.162)	–	–	–	–	711.8005/365
Band+PL	18.485(1.341)	−0.402(0.117)	−2.247(0.069)	308.928(20.511)	1.907(0.381)	−1.971(0.073)	–	–	671.8036/363
Band+BB	15.573(0.758)	−1.318(0.046)	−2.241(0.037)	1127.925(361.386)	–	–	2.463(0.511)	51.778(2.662)	675.3321/363
Band+PL+BB	9.550(1.913)	−0.681(0.235)	−2.319(0.093)	527.331(132.426)	2.082(0.399)	−1.940(0.045)	4.70(2.02)$^{\uparrow 3\%}$	38.331(5.361)	664.4058/361
CPL	15.203(0.391)	−1.042(0.020)	–	513.199(32.800)	–	–	–	–	1216.3942/366
CPL+PL	15.608(0.653)	−0.545(0.089)	–	371.144(20.524)	2.166(0.199)	−1.867(0.014)	–	–	697.1201/364
CPL+PL+BB	8.080(1.179)	−0.678(0.189)	–	613.839(100.475)	2.476(0.291)	−1.883(0.016)	5.255(1.940)	38.762(4.014)	677.3680/362
Episode IV: 55.296 s − 60.992 s									
PL	–	–	–	–	4.510(0.204)	−1.777(0.030)	–	–	602.1991/349
PL+BB	–	–	–	–	2.105(0.214)	−1.8790(0.0405)	8.371(1.799)	40.709(2.532)	361.4057/347
SBPL	8.975(0.422)	−0.9403(0.0109)	−2.168(0.059)	124.858(26.126)	–	–	–	–	338.5147/347
Band	16.029(3.379)	−0.764(0.136)	−2.158(0.059)	210.044(41.491)	–	–	–	–	338.3333/347
CPL	13.696(1.923)	−0.852(0.101)	–	260.946(41.210)	–	–	–	–	375.2517/348

A.2 GRB150210A

Table 3: Fitted parameters for time-integrated and time-resolved spectral analysis of GRB150210A. The values in parenthesis are the uncertainty in the measured parameter. The BEST models are highlighted with a light gray strip. The marked parameters in red indicate the values that are just beyond the error criteria limit. The percentage above threshold is indicated in superscript for reference. The models not included here either did not converge, or produced errors much higher than the threshold criteria.

GRB150210A									
	SBPL/Band/CPL				PL		BB		
	Amplitude [$\times 10^{-3}$] [ph cm^{-2} s^{-1} keV^{-1}]	α	β	Break/Peak Energy [keV]	Amplitude [$\times 10^{-3}$] [ph cm^{-2} s^{-1} keV^{-1}]	α	Amplitude [$\times 10^{-9}$] [ph cm^{-2} s^{-1} keV^{-1}]	kT [keV]	C-Stat/DOF
Time integrated: 0.064 s – 31.360 s									
PL	–	–	–	–	5.577(0.078)	−1.722(0.011)	–	–	4524.6274/463
PL+BB	–	–	–	–	2.701(0.107)	−1.885(0.024)	150.0(11.4)	161.504(3.890)	1181.2611/461
SBPL	7.084(0.094)	−1.109(0.013)	−3.002(0.075)	2407.710(234.738)	–	–	–	–	559.6879/461
SBPL+PL	6.213(0.257)	−1.037(0.030)	−4.871(1.385)	3500.320(825.941)	0.585(0.015)	−1.883(0.040)	–	–	535.6464/459
SBPL+BB	6.799(0.115)	−1.173(0.019)	−2.689(0.141)	1269.2(559.9)$^{\dagger 4.1\%}$	–	–	0.847(0.327)	627.849(47.309)	536.1976/459
Band	7.501(0.108)	−1.050(0.016)	−2.992(0.080)	2901.180(244.238)	–	–	–	–	586.4775/461
Band+PL	6.104(0.229)	−0.875(0.037)	−4.372(2.860)	2546.090(192.528)	0.905(0.127)	−1.934(0.031)	–	–	549.1145/459
Band+BB	7.067(0.165)	−1.152(0.025)	−2.793(0.103)	2314.060(483.567)	–	–	1.460(0.454)	524.439(54.838)	538.2578/459
CPL	7.494(0.105)	−1.050(0.015)	–	2956.130(219.730)	–	–	–	–	655.9387/462
CPL+PL	6.103(0.228)	−0.875(0.036)	–	2545.850(181.029)	0.906(0.126)	−1.934(0.030)	–	–	549.3083/460
CPL+BB	7.007(0.149)	−1.158(0.024)	–	2687.350(351.752)	–	–	1.630(0.548)	493.875(50.835)	611.4883/460
Episode EX–A: −0.384 s – 0.836 s									
PL	–	–	–	–	34.902(0.618)	−1.424(0.012)	–	–	3284.5852/455
PL+BB	–	–	–	–	13.340(0.759)	−1.598(0.040)	155.960(12.750)	313.042(8.285)	1009.2582/453
SBPL	30.180(0.771)	−0.778(0.021)	−3.726(0.256)	2587.658(228.090)	–	–	–	–	469.6516/453
Band	32.499(0.827)	−0.645(0.026)	−3.838(0.323)	2676.717(143.650)	–	–	–	–	468.1126/453
CPL	32.479(0.826)	−0.648(0.025)	–	2708.581(134.388)	–	–	–	–	470.3056/454
Episode I: 0.064 s – 0.836 s									
PL	–	–	–	–	52.282(0.924)	−1.400(0.011)	–	–	3575.1094/455
PL+BB	–	–	–	–	20.832(0.994)	−1.562(0.033)	128.383(9.017)	388.044(8.892)	1064.7013/453
SBPL	43.795(1.096)	−0.757(0.020)	−3.743(0.246)	2671.030(219.973)	–	–	–	–	468.2953/453
Band	47.127(1.173)	−0.624(0.024)	−3.912(0.321)	2784.240(136.617)	–	–	–	–	467.6772/453
CPL	47.103(1.171)	−0.626(0.023)	–	2817.800(127.954)	–	–	–	–	470.1885/454
Episode II: 0.832 s – 2.176 s									
PL	–	–	–	–	46.711(0.624)	−1.537(0.009)	–	–	7117.1362/463
PL+BB	–	–	–	–	14.841(0.712)	−1.722(0.030)	252.004(13.824)	306.825(5.527)	1240.0648/461
SBPL	41.202(0.824)	−0.726(0.017)	−3.065(0.089)	1921.550(112.148)	–	–	–	–	512.7466/461
SBPL+PL	35.793(1.547)	−0.634(0.032)	−3.166(0.171)	1832.850(151.070)	2.844(0.714)	−1.869(0.076)	–	–	494.9419/459
SBPL+BB	37.599(1.475)	−0.840(0.041)	−3.127(0.104)	2480.584(285.713)	–	–	75.3(30.4)$^{\dagger 0.3\%}$	273.699(28.994)	504.6757/459
Band	44.647(0.887)	−0.605(0.021)	−3.101(0.099)	2523.990(110.447)	–	–	–	–	532.8586/461
Band+PL	36.291(1.493)	−0.432(0.038)	−3.378(0.291)	2310.260(104.513)	4.195(0.681)	−1.876(0.056)	–	–	488.0091/459
Band+BB	40.320(1.361)	−0.789(0.039)	−3.061(0.109)	3054.780(242.188)	–	–	41.275(12.832)	366.519(28.805)	507.9320/459
CPL	44.549(0.879)	−0.620(0.019)	–	2670.640(94.733)	–	–	–	–	561.2298/462
CPL+PL	36.827(1.473)	−0.470(0.033)	–	2472.580(83.394)	4.077(0.646)	−1.844(0.044)	–	–	493.7455/460
CPL+BB	40.140(1.283)	−0.791(0.035)	–	3177.840(199.662)	–	–	44.614(12.819)	355.405(27.218)	536.0486/460
Episode III: 2.176 s – 3.328 s									
PL	–	–	–	–	26.346(0.535)	−1.555(0.016)	–	–	1949.4388/458
PL+BB	–	–	–	–	9.601(0.795)	−1.748(0.050)	1243.750(131.947)	134.264(4.270)	729.5074/456
SBPL	29.700(0.767)	−0.894(0.034)	−2.610(0.112)	839.092(111.306)	–	–	–	–	541.8218/456
Band	33.648(1.149)	−0.797(0.042)	−2.641(0.124)	1311.920(162.680)	–	–	–	–	540.5592/456
CPL	33.093(1.001)	−0.818(0.036)	–	1449.000(152.285)	–	–	–	–	570.6387/457
Episode IV: 3.328 s – 17.344 s									
PL	–	–	–	–	2.993(0.115)	−1.826(0.028)	–	–	669.7643/459
PL+BB	–	–	–	–	1.668(0.164)	−1.956(0.050)	4850.469(1189.531)	37.280(2.481)	518.1394/457
SBPL	4.693(0.204)	−1.285(0.083)	−2.225(0.079)	134.260(33.788)	–	–	–	–	488.5370/457
Band	7.047(1.228)	−1.156(0.107)	−2.216(0.080)	203.470(46.043)	–	–	–	–	488.5068/457
CPL	5.653(0.623)	−1.283(0.076)	–	308.256(70.069)	–	–	–	–	510.4138/458
Episode V: 17.344 s – 31.360 s									
PL	–	–	–	–	1.309(0.112)	−1.970(0.057)	–	–	533.1559/462
CPL	3.571(1.194)	−1.325(0.190)	–	121.190(32.946)	–	–	–	–	504.2606/461